

NANYANG TECHNOLOGICAL UNIVERSITY

MIDTERM II

MH2100 – Calculus III

November 2019

TIME ALLOWED: 90 minutes

Name:

Matric. no.:

Tutor group:

INSTRUCTIONS TO CANDIDATES

1. **DO NOT TURN OVER PAPER UNTIL INSTRUCTED.**
2. This midterm paper contains **THREE (3)** questions.
3. Answer **ALL** questions. The marks for each question are indicated at the beginning of each question.
4. Candidates can write anywhere on this midterm paper.
5. **One** two-sided A4 piece of paper with handwritten notes is permitted.

QUESTION 1.**(32 marks)**

- (i) Use Lagrange Multipliers to find the maximum value of the function

$$f(x, y, z) = 14x + 8y + 12z$$

subject to the constraint $x^2 + y^2 + z^2 = 101$.

- (ii) Evaluate the iterated integral.

$$\int_{-1}^1 \int_0^1 ye^{xy} dx dy$$

- (iii) Find the volume of the solid bounded by the surface $z = 5 + (x-4)^2 + 2y$ and the planes $x = 0$, $y = 0$, $z = 0$, $x = 3$, $y = 4$.

- (iv) Evaluate the iterated integral by converting to polar coordinates.

$$\int_0^{\sqrt{\pi}} \int_0^{\sqrt{\pi-y^2}} \sin(x^2 + y^2) dx dy$$

- (v) Use cylindrical coordinates to evaluate

$$\iiint_E y dV,$$

where E is the solid that lies between the cylinders $x^2 + y^2 = 3$ and $x^2 + y^2 = 7$, above the xy -plane and below the plane $z = x + 4$.

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QUESTION 2.**(34 marks)**

- (i) Use spherical coordinates to evaluate

$$\iiint_B \sqrt{x^2 + y^2 + z^2} dV,$$

where B is the ball $x^2 + y^2 + z^2 \leq 6$.

- (ii) Find the area of the region bounded by the curves $y = \sqrt{x+1}$ and $x = 3y - 1$.

- (iii) Evaluate the iterated integral

$$\int_0^{\sqrt{\pi/2}} \int_y^{\sqrt{\pi/2}} \cos(x^2) dx dy.$$

- (iv) Use the transformation $x = \frac{u}{v}$; $y = v$ to evaluate the integral

$$\iint_R xy^3 dA,$$

where R is the region in the first quadrant ($x \geq 0$ and $y \geq 0$) bounded by the lines $y = x$, $y = 3x$, and the hyperbolas $xy = 1$ and $xy = 4$.

- (v) Find the volume of the solid inside the cylinder $x^2 + y^2 = 16$ and the ellipsoid $2x^2 + 2y^2 + z^2 = 32$.

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QUESTION 3.**(34 marks)**

- (i) Evaluate the line integral

$$\int_C xy ds,$$

where C is the line segment from $(-2, -1)$ to $(4, 5)$.

- (ii) Find the work done by the force field
- $\mathbf{F}(x, y) = y\mathbf{i} - x\mathbf{j}$
- on a particle that moves along the curve
- $x = 2y^3 - 6y^2 + 1$
- from
- $(-3, 1)$
- to
- $(1, 3)$
- .

- (iii) Evaluate the line integral

$$\int_C x^2 y \cos z ds,$$

where C has parametrisation $x = \cos t$, $y = \sin t$ and $z = t$, with $0 \leq t \leq \pi$.

- (iv) Let
- $\mathbf{F}(x, y) = y\mathbf{i} + (x + 10y)\mathbf{j}$
- and let
- C
- be the upper semicircle that starts at
- $(1, 2)$
- and ends at
- $(5, 2)$
- .

(1) Find a function f such that $\mathbf{F} = \nabla f$.(2) Hence, or otherwise, evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$.

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